

7 Meiosis increases genetic variation through independent assortment and crossing over.

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(a) (i) In which stage of meiosis does independent assortment occur?

(1)

- ☐ A metaphase I
- ☐ B metaphase II
- ☐ C telophase I
- ☐ D telophase II

(ii) In which stage of meiosis does crossing over occur?

(1)

- ☐ A metaphase I
- ☐ B metaphase II
- ☐ C prophase I
- ☐ D prophase II



- (b) Genetic variation increases due to the production of new combinations of alleles.

A cell has gene A and gene B with alleles A, a and B, b.

Meiosis could produce the following percentages of gametes containing the combinations of alleles shown in the table.

Combination of alleles	Percentage of total gametes produced (%)
AB	48
Ab	2
aB	2
ab	48

Explain why the percentages are not 25% for each combination of alleles.

(3)

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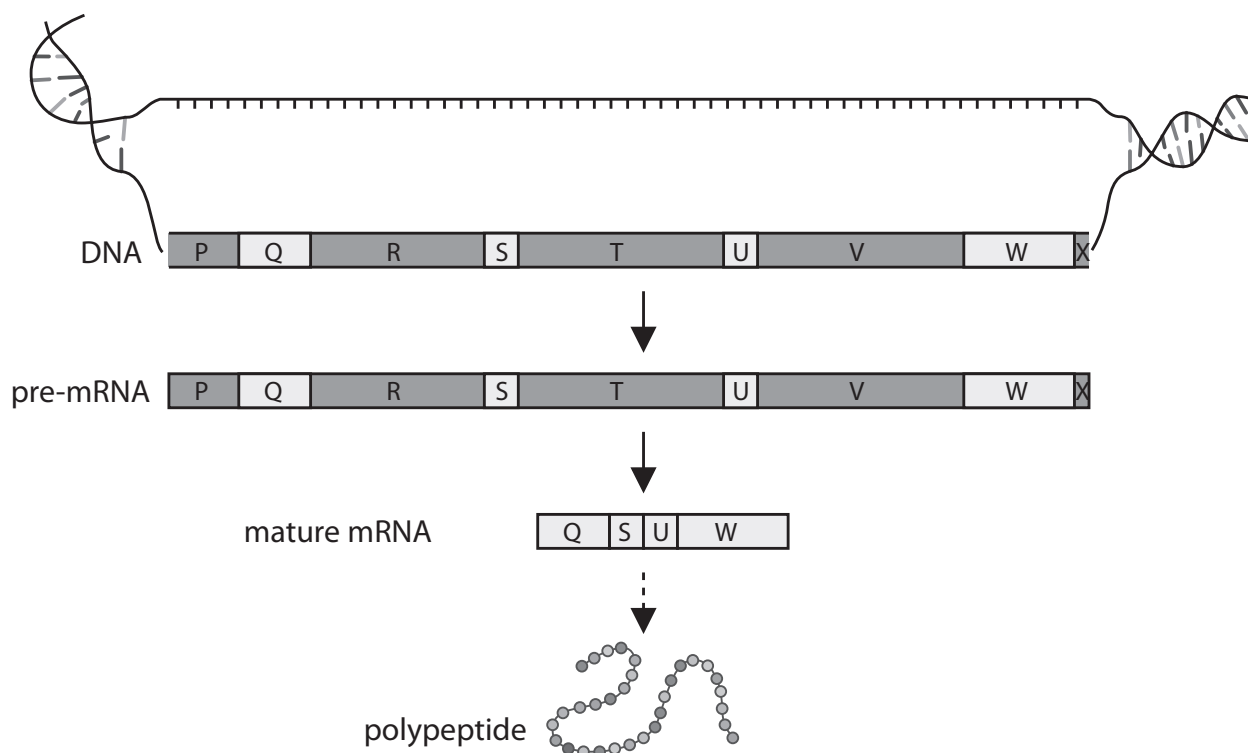
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- (c) More than one type of protein can be synthesised from the RNA produced from one gene.

The diagram shows parts of this process.



Describe how more than one type of protein can be synthesised from the RNA produced from one gene.

Use the information in the diagram to support your answer.

(5)

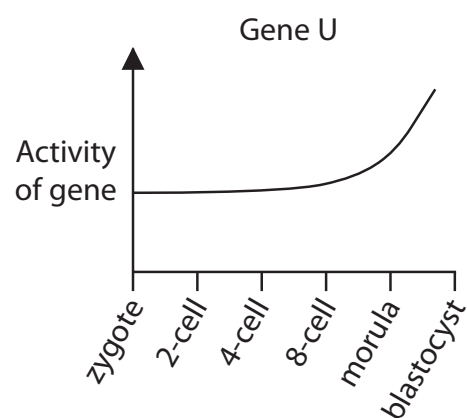
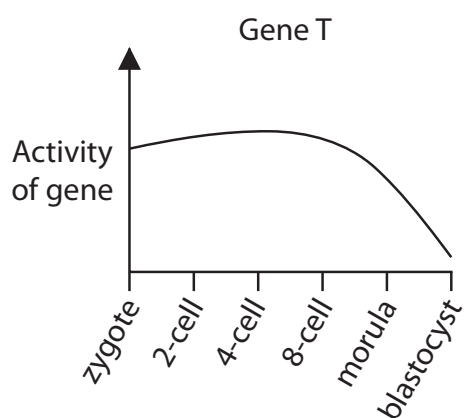


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- (d) The graphs show the activity of two genes, T and U, as a mouse zygote (fertilised egg cell) develops into a blastocyst.



Comment on the changes in the activity of these two genes.

(3)

(Total for Question 7 = 13 marks)

